

CS 340

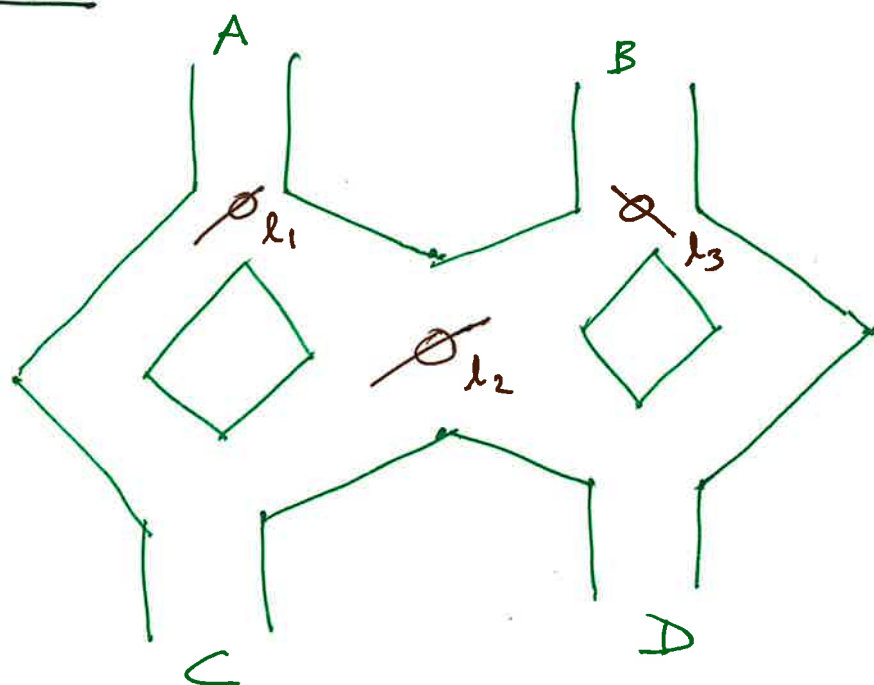
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Lecture 4

Deterministic Finite Automata Contd...

In this lecture, we first model a system using a DFA, which is more complex than the systems considered in the last lecture. After this, we discuss a few closure properties of the class of languages accepted by DFA.

Example # (Modeling a toy)



l_1, l_2, l_3
are levers

A marble is dropped from either A or B,
depending on positions of the levers it exits
either C or D. Levers reverse their
orientation after the marble passes through them.

This can be modeled as a finite state system
(DFA). The state is given by positions of the levers.

we represent  state of lever by 0
and  " by 1

For instance, state in figure shown above is

$\downarrow l_2$
(1 | 0)
 $\uparrow \quad \uparrow$
 $l_1 \quad l_3$

State-0 (of lever) provides
passage to its right side,
state-1 provides passage to
its left.

It may also be noted that if l_2 is in state 0
the ~~only~~ marble hitting l_2 exits at D and if l_2
is in state-1 then the marble hitting l_2 exits at C.

Transition fn may be represented by a table

Input ↓ state	A	B
000	11 0/D	00 1/D
001	11 1/D	01 0/D
010	10 0/C	01 1/D
011	10 1/C	01 1/D
100	00 0/C	00 0/C
101	00 1/C	10 1/D
110	01 0/C	11 0/D
111	01 1/C	11 1/D
		10 0/C

Entries in the table are in state/output format

The toy may be seen as taking ~~an~~ inputs over $\{A, B\}^*$.
 Input 'A' means dropping a marble on the left tunnel and
 input 'B' means dropping a marble on the right tunnel.
 Let us look at the output produced by input AABA
 starting in state '000'.



Let us define acceptance for a string $x \in \{A, B\}^*$
 as the exit of the last marble at 'C'. More formally,
 let $L = \left\{ x \in \{A, B\}^* \mid \begin{array}{l} \text{input } x \text{ causes last} \\ \text{marble in the toy} \\ \text{to exit at C} \end{array} \right\}$

Exercise: show that L is accepted by a DFA. Design
 this DFA.

Solution Design a DFA with states of the form
 ijk/α , where $i, j, k \in \{0, 1\}$ and $\alpha \in \{C, D\}$

$$\delta(ijk/\alpha, \frac{A}{B}) = \text{table}(ijk, \frac{A}{B})$$

(Note that ~~new~~ state/output does not
 depend on α)

['table' is the
 table shown
 on the last
 page]

Make states of the form ijk/C as
 accepting. Take initial state as $000/D$.

□

Closure Properties

We first show that the class of languages recognized by DFA are closed under (set theoretic) union, intersection and complement.

More formally let

$$R = \{ L \subseteq \Sigma^* \mid \text{there is a DFA } M \text{ s.t. } L = L(M) \}.$$

For any $L_1, L_2 \in R$, we show that

$$L_1 \cup L_2, L_1 \cap L_2, \bar{L}_1 \in R$$

(Here $\bar{L}_1$ is complement of L_1 in Σ^* , that is $\bar{L}_1 = \Sigma^* - L_1$).

Theorem If $L_1, L_2 \in R$ then $L_1 \cup L_2 \in R$.

Proof:

Let $M_i = (Q_i, \Sigma, q_{0i}, \delta_i, F_i)$, for $i \in \{1, 2\}$

be s.t. $L_i = L(M_i)$.

Consider product automaton M designed as follows.

$$M = (Q_1 \times Q_2, \Sigma, (q_{01}, q_{02}), \delta, \overbrace{F_1 \times F_2}^F)$$

Where $\delta : (Q_1 \times Q_2) \times \Sigma \rightarrow Q_1 \times Q_2$ is defined as

$$\delta((q, q'), a) = (\delta_1(q, a), \delta_2(q', a))$$

$$\text{and } F = (Q_1 \times F_2) \cup (F_1 \times Q_2)$$

The construction above simulates both automata M_1 and M_2 in parallel in the first and second component of states of M . We accept the input if at least one of the components of the last state is final.

More precisely, it is easy to see that for $x \in \Sigma^*$

$$\delta((q_{01}, q_{02}), x) = (\delta_1(q_{01}, x), \delta_2(q_{02}, x)) \quad \text{--- (I)}$$

using (I), we see $L(M) = L(M_1) \cup L(M_2)$ as follows.

' $\subseteq$ ' let $x \in L(M)$

$\Rightarrow \delta_1(q_{01}, x) \in F_1$ or $\delta_2(q_{02}, x) \in F_2$ (or both)

w.l.o.g. let us assume

$$\delta_1(q_{01}, x) \in F_1$$

$$\Rightarrow x \in L(M_1)$$

$$\Rightarrow x \in L(M_1) \cup L(M_2)$$

' $\supseteq$ ' let $x \in L(M_1) \cup L(M_2)$

w.l.o.g. let us assume $x \in L(M_1)$

$$\Rightarrow \delta_1(q_{01}, x) \in F_1$$

$$\Rightarrow \delta((q_{01}, q_{02}), x) = (\delta_1(q_{01}, x), \delta_2(q_{02}, x))$$

$$\in F_1 \times Q_2 \subseteq F$$

$$\Rightarrow x \in L(M)$$

This proves both directions of the containment.

It remains to prove (I). This is done by induction on $|x|$

Base Case: $|x| = 0$, this implies $x = \varepsilon$

$$\begin{aligned}\delta(q_{01}, q_{02}, \varepsilon) &= (q_{01}, q_{02}) && [\text{By defn of } \hat{\delta}] \\ &= (\delta_1(q_{01}, \varepsilon), \delta_2(q_{02}, \varepsilon))\end{aligned}$$

Induction step: $x = ya$ ($y \in \Sigma^*$, $a \in \Sigma$)

$$\begin{aligned}\delta((q_{01}, q_{02}), x) &= \delta((q_{01}, q_{02}), ya) \\ &= \delta(\delta((q_{01}, q_{02}), y), a) && [\text{By defn of } \hat{\delta}] \\ &= \delta((\delta_1(q_{01}, y), \delta_2(q_{02}, y)), a) && [\text{By I.H.}] \\ &= (\delta_1(\delta_1(q_{01}, y), a), \delta_2(\delta_2(q_{02}, y), a)) \\ &= (\delta_1(q_{01}, ya), \delta_2(q_{02}, ya)) \\ &= (\delta_1(q_{01}, x), \delta_2(q_{02}, x)) \quad \square\end{aligned}$$

Remarks

Details of proofs of (I) and $LCM = L(M_1) \cup L(M_2)$ are straightforward. In future, we will omit such details for the reader to work out himself/herself.

Theorem If $L_1, L_2 \in R$ then $L_1 \cap L_2 \in R$.

Proof: Similar to the previous theorem, except set of final states F is taken as $F_1 \cap F_2$ in the construction of M . $\square$

Theorem If $L \in R$ then $\bar{L} \in R$.

Proof

Let $L = L(M)$, where

$$M = (Q, \Sigma, q_0, \delta, F)$$

We define $M' = (Q, \Sigma, q_0, \delta, Q - F)$

$$x \in L(M') \iff \delta(q_0, x) \in Q - F$$

$$\iff \delta(q_0, x) \notin F$$

$$\iff x \notin L(M)$$

$$\iff x \in \overline{L(M)}$$

$$\Rightarrow L(M') = \overline{L(M)} = \bar{L}$$

$\square$

Let us consider a few ~~at~~ more operations on languages which are not just set theoretic.

$$L_1 \cdot L_2 = \{x_1 x_2 \mid x_1 \in L_1, x_2 \in L_2\}$$

$$L^* = \{x_1 \dots x_n \mid \exists n \geq 0 \text{ s.t. } x_1, x_2, \dots, x_n \in L\}$$



Kleene closure
of L

Set R is closed under these operations also, we will show this after a couple of lectures, when we have some more tools. Following closure results we can show we can show today itself.

Define

$$\text{pre}(L) = \{x \in \Sigma^* \mid \exists y \in \Sigma^* (xy \in L)\}$$

$$\text{suff}(L) = \{x \in \Sigma^* \mid \exists y \in \Sigma^* (yx \in L)\}$$

Let $M = (Q, \Sigma, q_0, \delta, F)$

define $M_{\text{pre}} = (Q, \Sigma, q_0, \delta, F')$

where $F' = \{q \in Q \mid \exists y \in \Sigma^* (\delta(q, y) \in F)\}$

Exercise: show that $L(M_{\text{pre}}) = \text{pre}(L(M))$

What can you say about $\text{suff}(L)$, for $L \in R$?